Solution

Explore Sinusoids

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Set up workspace

```
clear
close all
```

Problem 1

As frequency increases, the rate of oscillations increase, which is equivalent to the period decreasing. Doubling the frequency cuts the period in half. Sinusoids with distinct frequencies appear distinct.

The phase shifts the zero crossings of the sinusoid (and all other values), and is equivalent to a shift in time.

```
t = -1:0.001:1;

% Part a

x = 5*cos(2*pi*t);

figure
plot(t,x)
xlabel('time (sec)')
ylabel('amplitude')
title('Part a, f = 1 Hz, \phi = 0 ')

% Part b

x = 5*cos(2*pi*t - pi/4);

figure
plot(t,x)
xlabel('time (sec)')
ylabel('amplitude')
title('Part b, f = 1 Hz, \phi = -\pi/4 ')

% Part c

x = 5*cos(2*pi*2*t - pi/4);

figure
plot(t,x)
```

```
xlabel('time (sec)')
ylabel('amplitude')
title('Part c, f = 2 Hz, \phi = -\pi/4 ')

% Part d

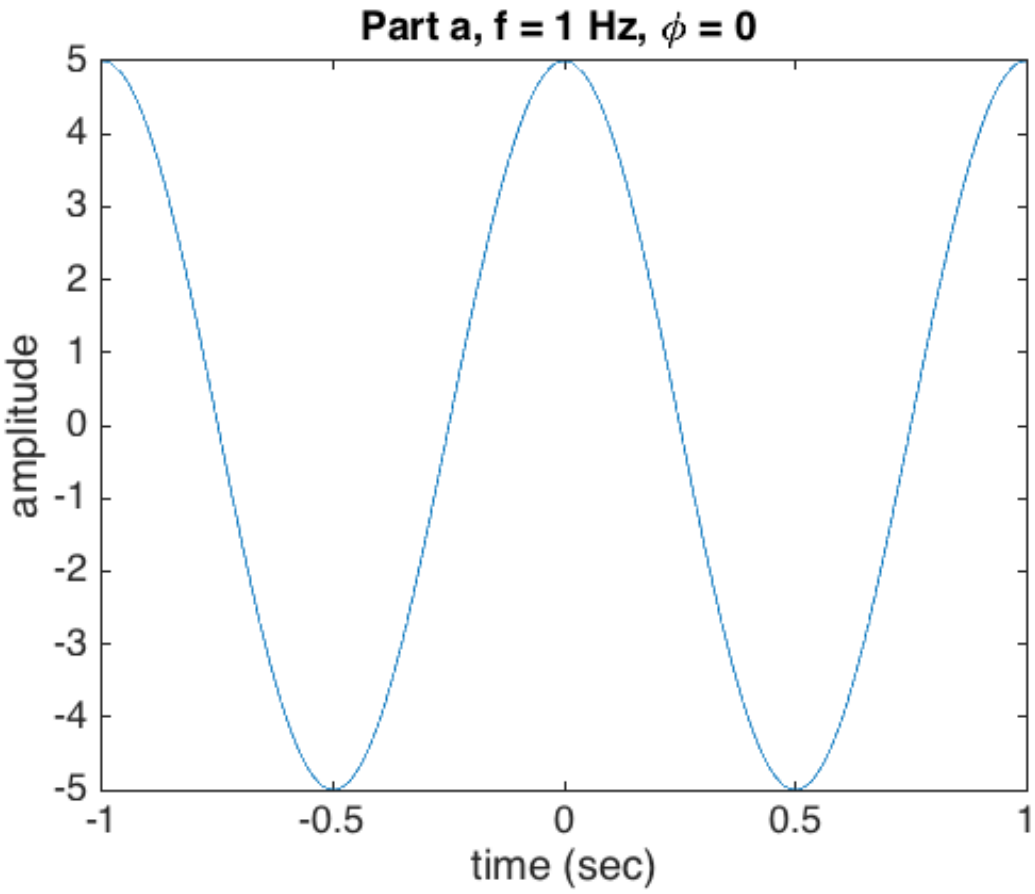
x = 5*cos(2*pi*5*t - pi/4);

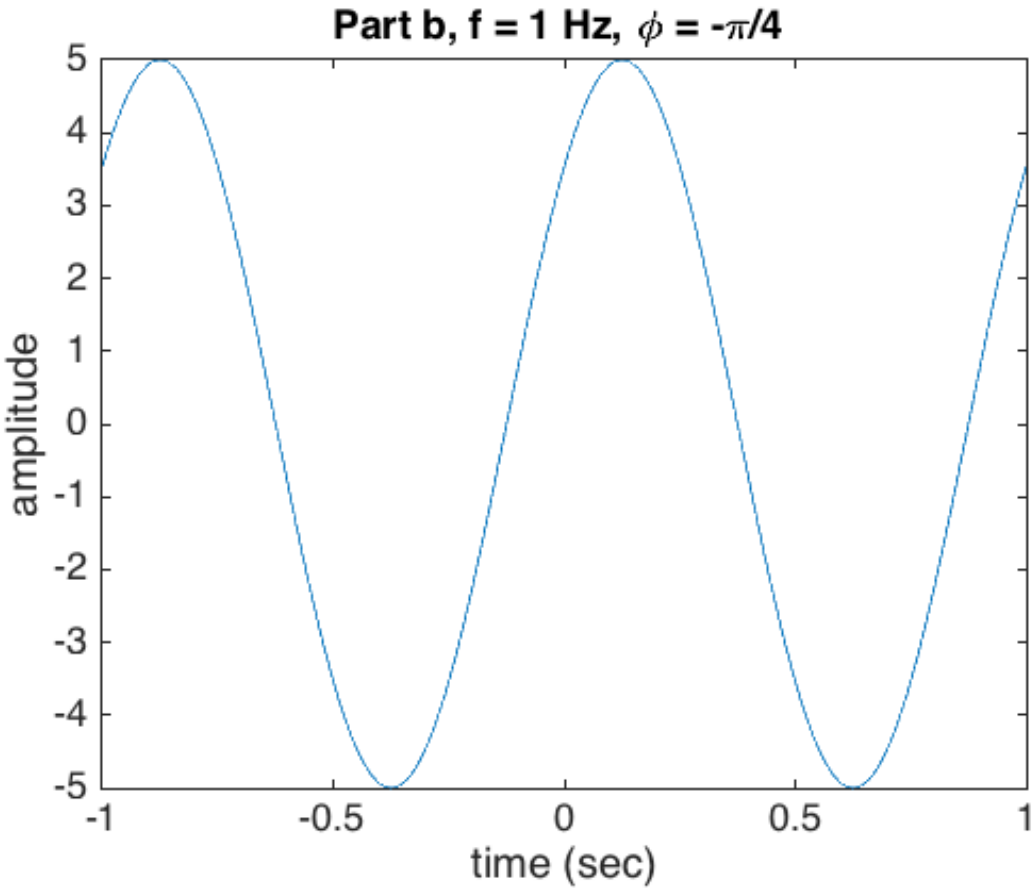
figure
plot(t,x)
xlabel('time (sec)')
ylabel('amplitude')
title('Part d, f = 5 Hz, \phi = -\pi/4 ')

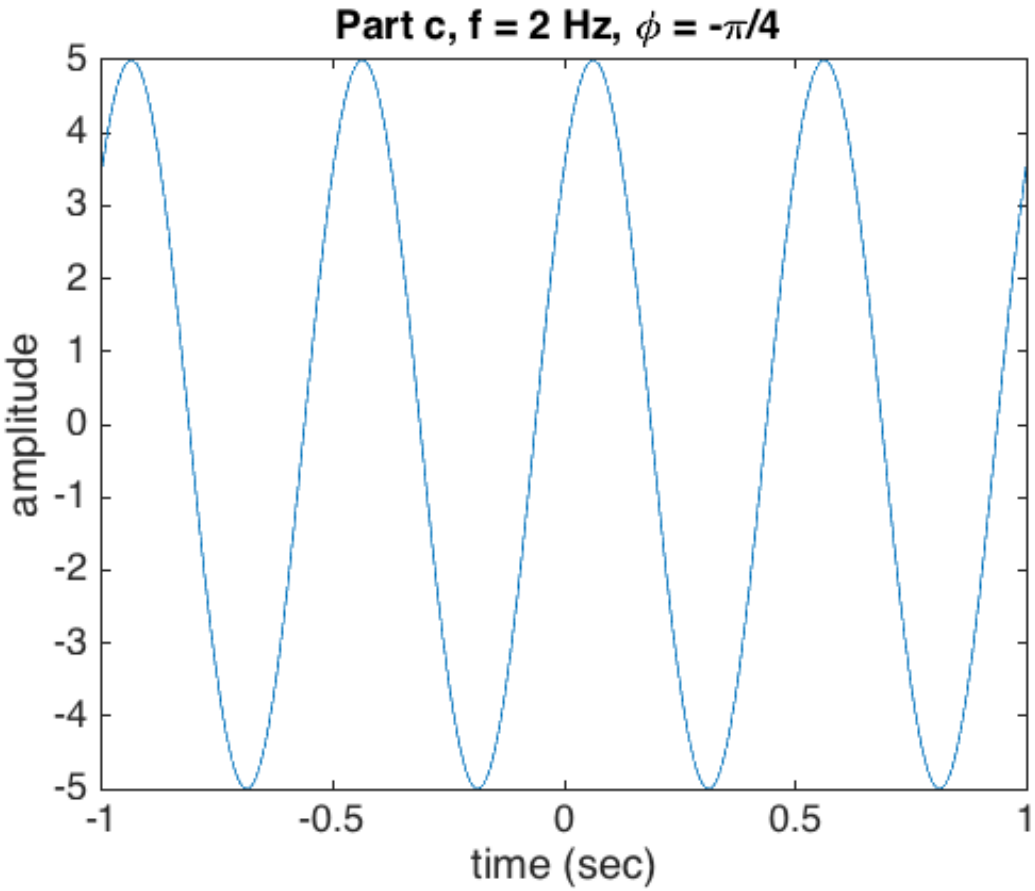
% Part e

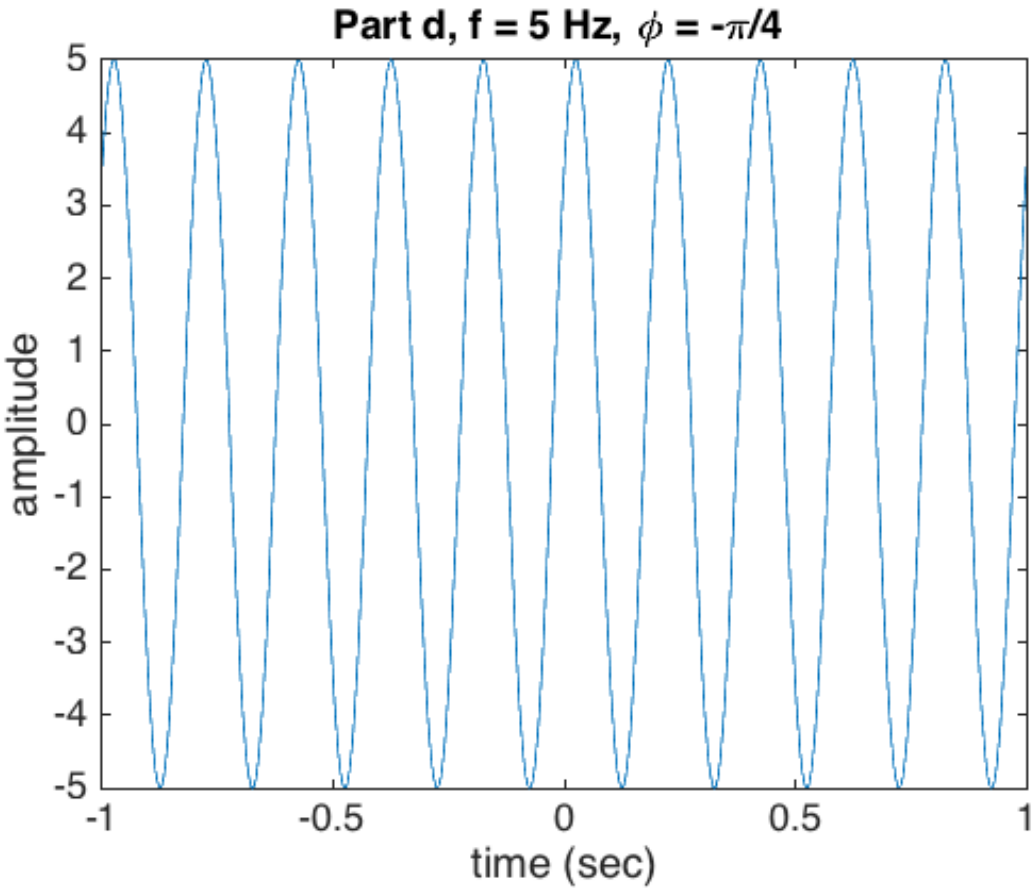
x = 5*cos(2*pi*10*t - pi/4);

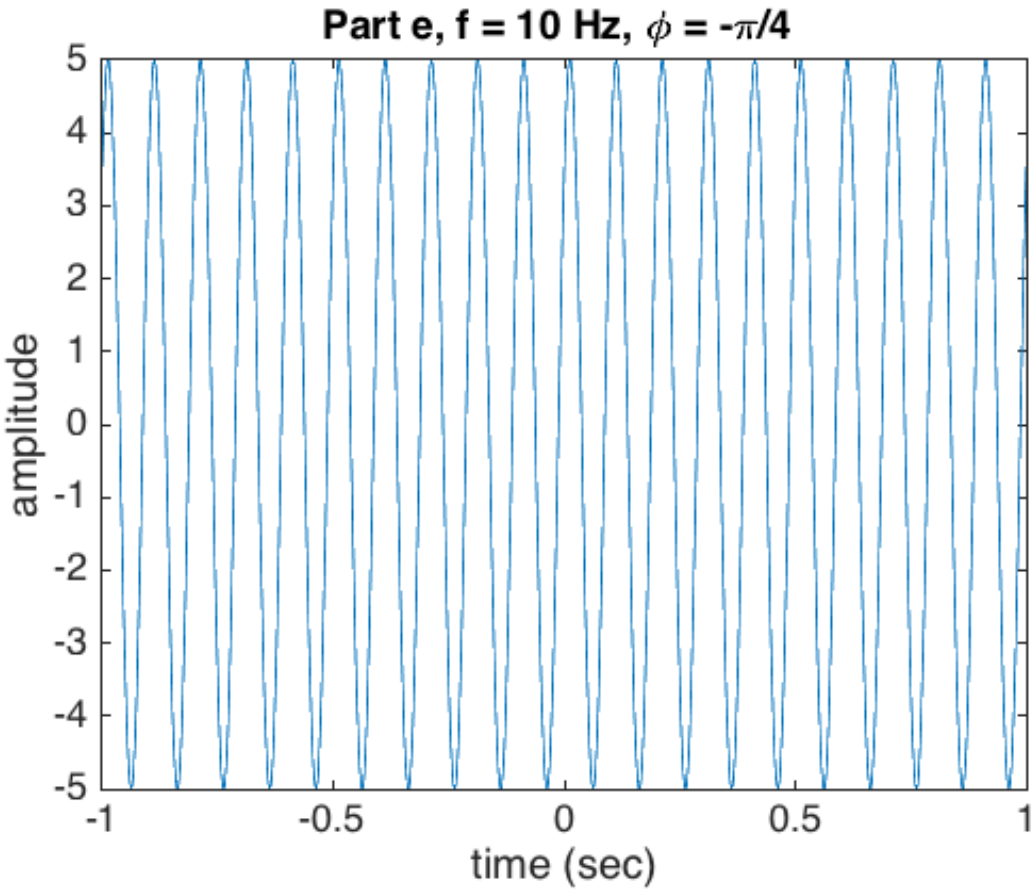
figure
plot(t,x)
xlabel('time (sec)')
ylabel('amplitude')
title('Part e, f = 10 Hz, \phi = -\pi/4 ')
```




Part b, f = 1 Hz, ϕ = -π/4
amplitude
time (sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
-1
-0.5
0
0.5
1


Part c, f = 2 Hz, ϕ = -π/4
amplitude
time (sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
-1
-0.5
0
0.5
1


Part d, f = 5 Hz, ϕ = -π/4
amplitude
time (sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
-1
-0.5
0
0.5
1


Part e, f = 10 Hz, ϕ = -π/4
amplitude
time (sec)
5
4
3
2
1
0
-1
-2
-3
-4
-5
-1
-0.5
0
0.5
1

Problem 2

As frequency increases, the rate of oscillations increase, that is, the period decreases, but only for a limited range of frequencies. Discrete time sinusoids are only unique within a 2pi range of frequencies. So parts b, d, and e result in identical signals because the frequencies are shifted by multiples of 2pi.

```
n = -10:90;

% Part a

x = 3*cos(pi*n/50 + pi/4);

figure
stem(n,x)
xlabel('n (samples)')
ylabel('Amplitude')
xlim([-10,90])
title('Part a, \Omega = \pi/50 rads, \phi = \pi/4 ')

% Part b

x = 3*cos(pi*n/5);

figure
stem(n,x)
xlabel('n (samples)')
ylabel('Amplitude')
title('Part b, \Omega = \pi/5 rads, \phi = 0 ')
xlim([-10,90])

% Part c

x = 3*cos(pi*n/2);

figure
stem(n,x)
xlabel('n (samples)')
ylabel('Amplitude')
title('Part c, \Omega = \pi/2 rads, \phi = 0 ')
xlim([-10,90])

% Part d

x = 3*cos(11*pi*n/5);

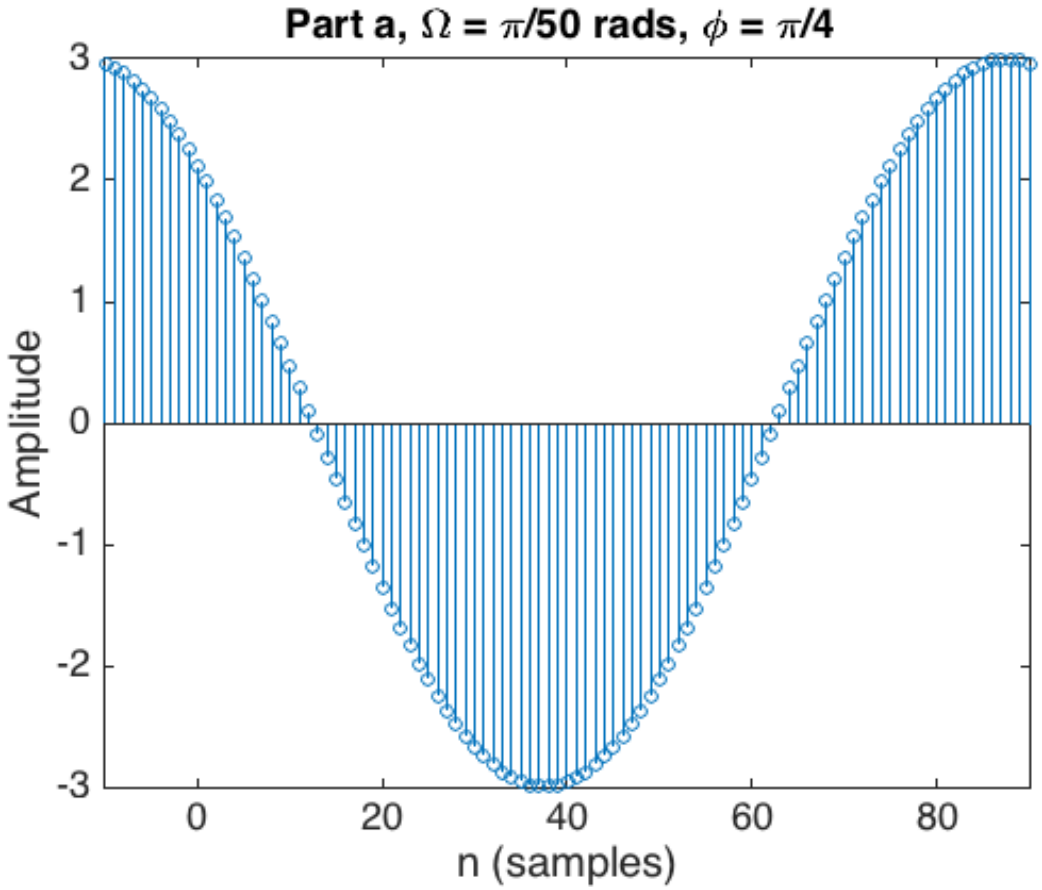
figure
stem(n,x)
xlabel('n (samples)')
ylabel('Amplitude')
title('Part d, \Omega = 11\pi/5 rads, \phi = 0 ')
xlim([-10,90])

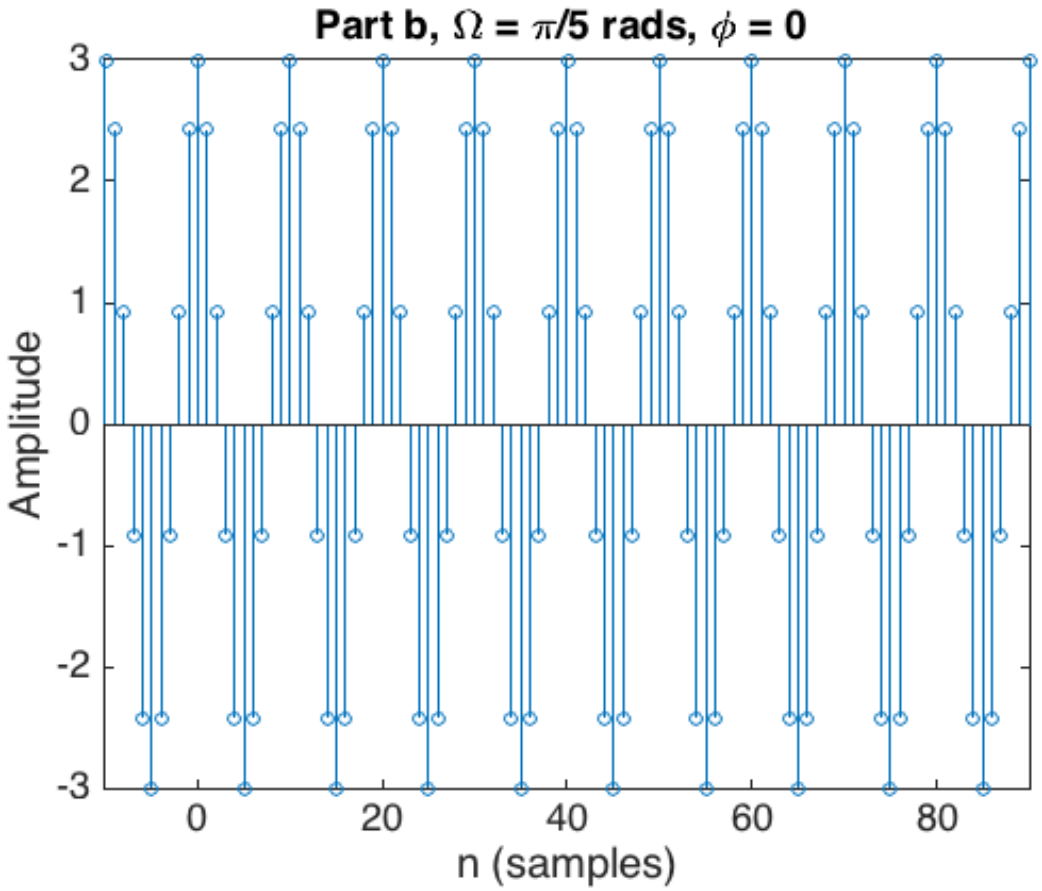
% Part e

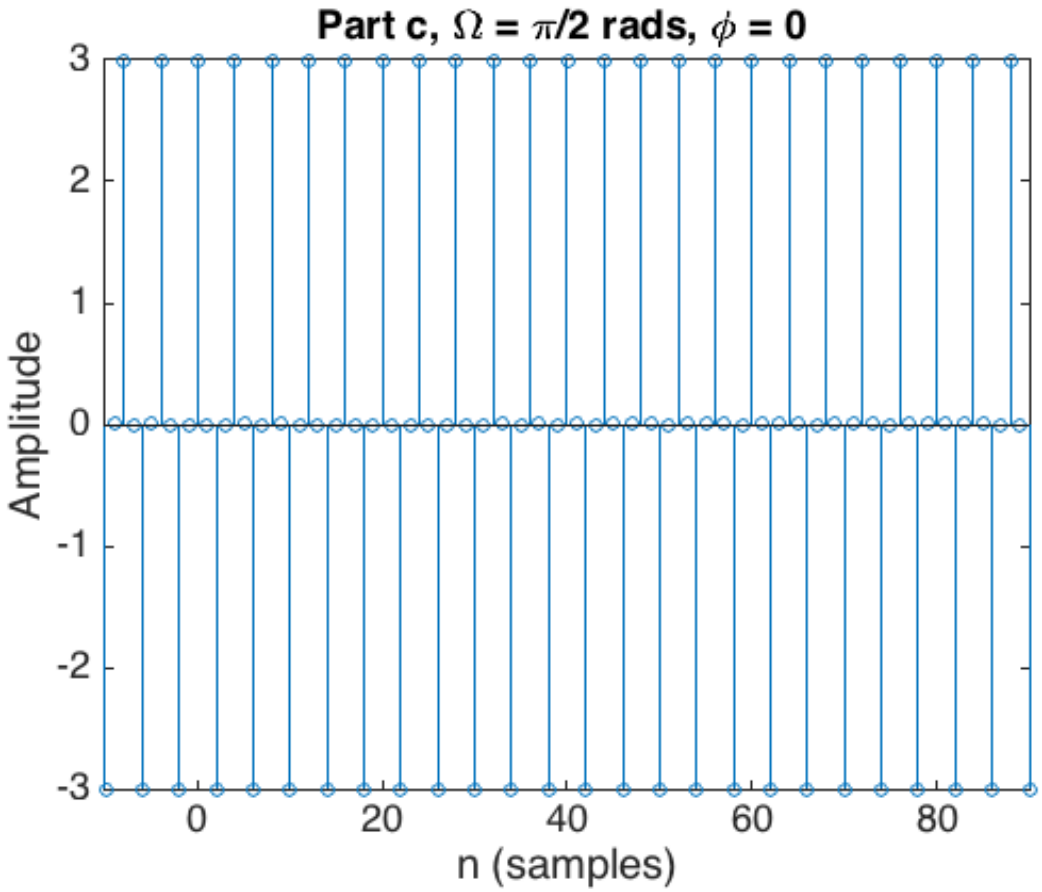
x = 3*cos(21*pi*n/5);

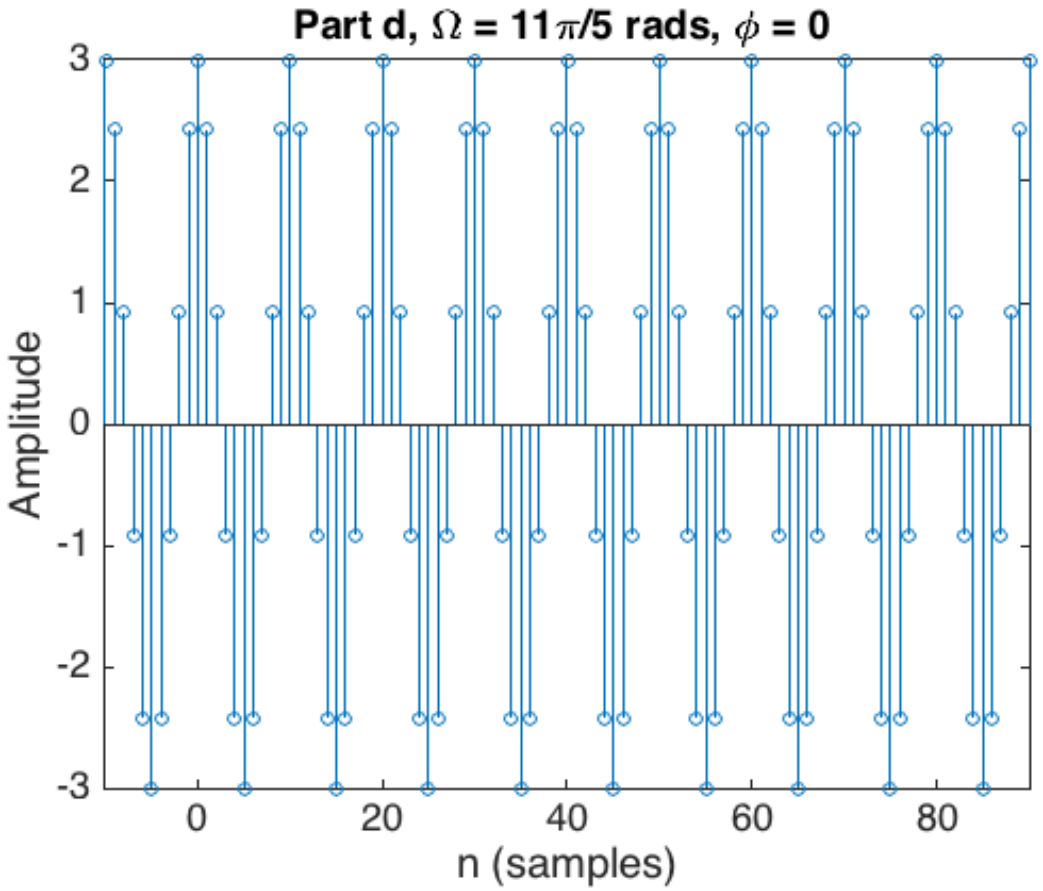
figure
stem(n,x)
```

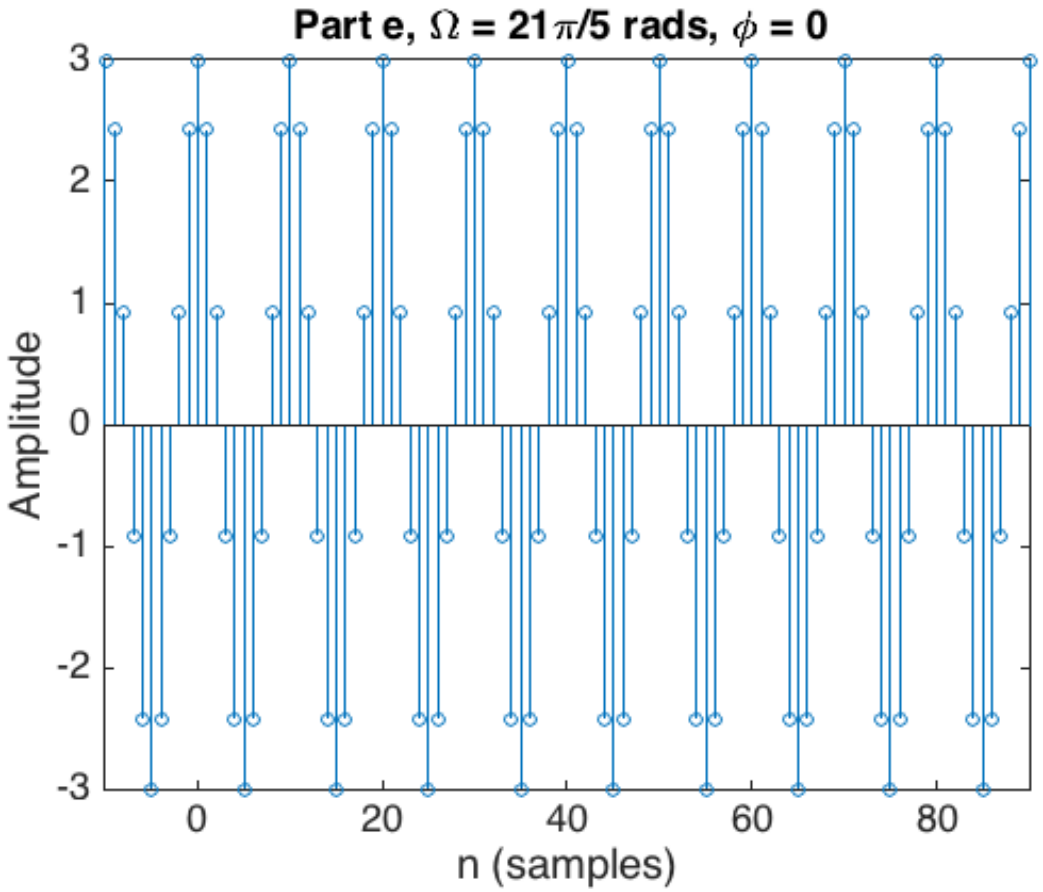
```
xlabel('n (samples)')
ylabel('Amplitude')
title('Part e, \Omega = 21\pi/5 rads, \phi = 0 ')
xlim([-10,90])
```




Part b, Ω = π/5 rads, ϕ = 0
Amplitude
n (samples)
3
2
1
0
-1
-2
-3
0
20
40
60
80


Part c, Ω = π/2 rads, ϕ = 0
Amplitude
n (samples)
3
2
1
0
-1
-2
-3
0
20
40
60
80


Part d, Ω = 11π/5 rads, ϕ = 0
Amplitude
n (samples)
3
2
1
0
-1
-2
-3
0
20
40
60
80


Part e, Ω = 21π/5 rads, ϕ = 0
Amplitude
n (samples)
3
2
1
0
-1
-2
-3
0
20
40
60
80

Problem 3

Both c and s have unit magnitude, so the product of c and s also has unit magnitude and phase given by the sum of the individual phases: 3pi + 6pi = 9pi.

```
t = 0:0.001:1;

j = sqrt(-1);

c = exp(j*3*pi*t);
s = exp(j*6*pi*t);

% Part a

c(50:54)

% Part b

y = 2*c + s;

figure
subplot(211)
plot(t,real(y))
title('Real part of y ')
ylabel('Amplitude')

subplot(212)
plot(t,imag(y))
title('Imaginary part of y ')
ylabel('Amplitude')
xlabel('Time (sec)')

% Part c

z = c.*s;

figure
subplot(211)
plot(t,abs(z))
title('Magnitude of c.*s ')
ylabel('Amplitude')
ylim([0 1.1])

subplot(212)
plot(t,angle(z))
title('Phase of c.*s ')
ylabel('Radians')
xlabel('Time (sec)')
```

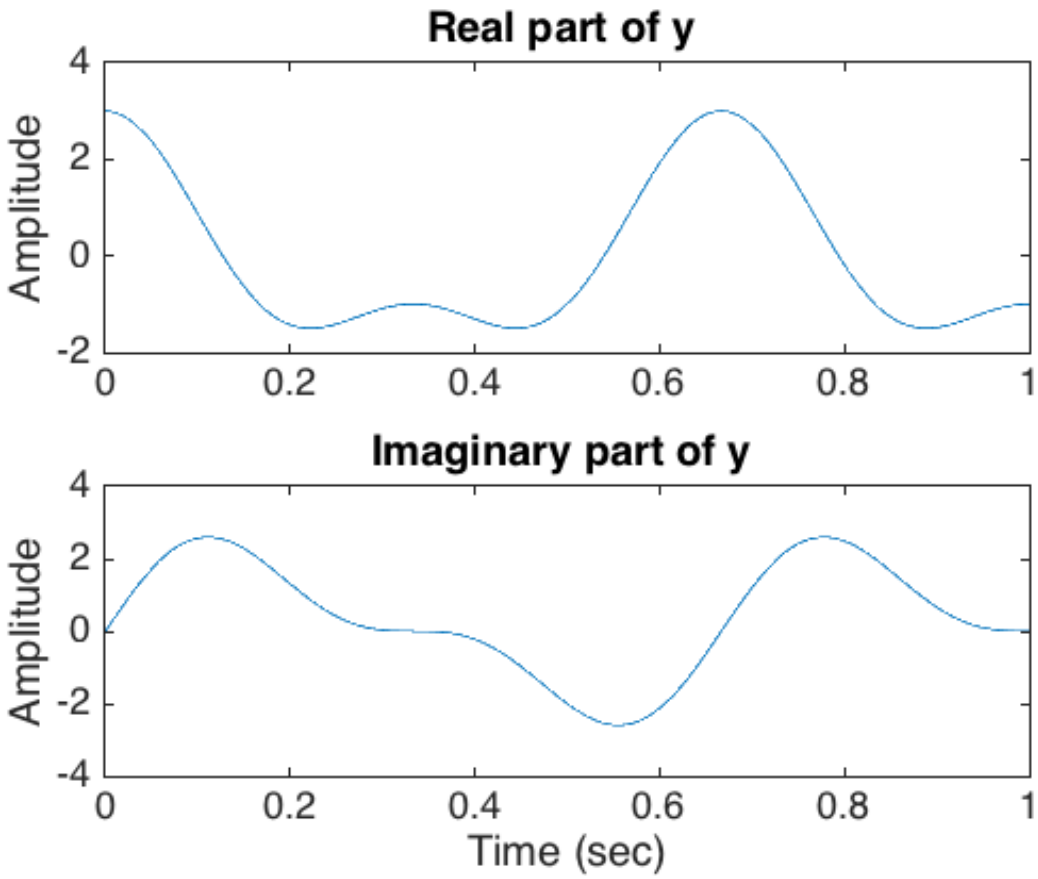
```
ans =

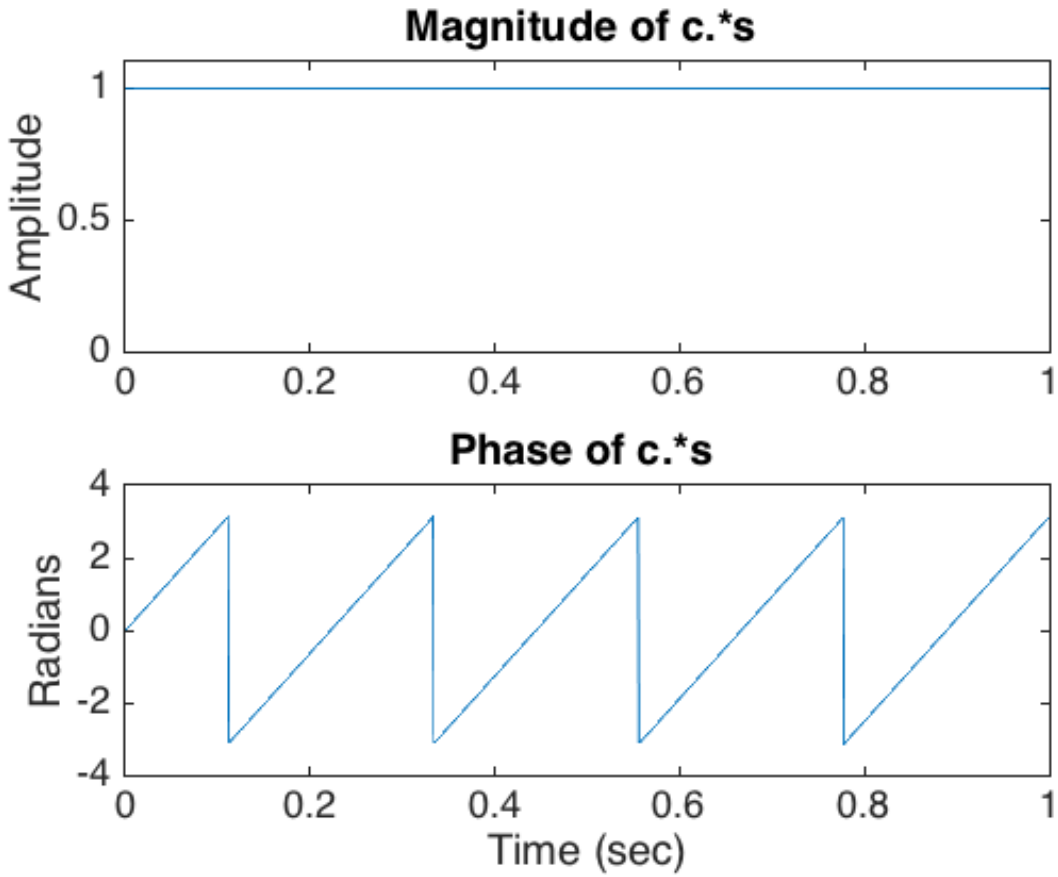
  Columns 1 through 4

   0.8952 + 0.4456i   0.8910 + 0.4540i   0.8867 + 0.4624i   0.8823 + 0.4707i

  Column 5
```

```
0.8778 + 0.4790i
```





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